

Enervision 360

Smart Energy & Infrastructure Management

Powered by AI & IoT — Built for the Buildings of Tomorrow



The Problem: Energy & Operations Are Getting Costly

⚡ Skyrocketing Energy Bills

Electricity costs continue to rise with no clear way to identify waste or optimize consumption across facilities.

💣 Unexpected Equipment Failures

Critical systems break down without warning, causing costly emergency repairs and damaging operational continuity.

📄 Manual & Inefficient Monitoring

Teams rely on manual checks and spreadsheets, wasting time while missing the real-time insights needed to act fast.

🔍 No Centralized Visibility

Each location operates in isolation — management has no single view of what's happening across their entire portfolio.



Introducing Enervision 360

Enervision 360 is an intelligent, always-on monitoring platform that connects your entire infrastructure — from HVAC systems to lighting and critical equipment — giving you real-time visibility, instant alerts, and AI-powered recommendations, all from a single dashboard.

Automatic Appliance Tracking

The system identifies every connected device without manual setup, continuously monitoring performance around the clock.

Real-Time Alerts

Instant notifications reach the right people the moment an anomaly is detected — before it becomes a breakdown.

Actionable Insights

Clear, plain-language recommendations help your team act decisively — no technical expertise required.



What Enervision 360 Does for Your Business



Identifies Appliances Automatically

Recognizes every connected device the moment it comes online — no manual inventory or tagging required.



Detects Faults Instantly

Spots irregularities in real time, flagging issues the moment they appear so your team can respond immediately.



Predicts Failures in Advance

AI models analyze patterns to forecast equipment problems days or weeks before they cause a disruption.



Optimizes Energy Usage

Continuously fine-tunes energy consumption across your facilities, cutting waste and lowering your utility bills.

How It Works – From Sensor to Smart Decision

The system uses industrial telemetry protocols and an edge-to-cloud architecture to move high-frequency equipment data from field devices into analytics pipelines in real time. Sensors stream at high sample rates over MQTT, Modbus, and BACnet transports via RS-485 or LoRaWAN, with edge gateways handling normalization, buffering, and secure forwarding into the cloud.



Edge Layer Specs

- Sampling rate up to 1kHz
- Supports 200+ sensor types
- Offline buffering up to 72hrs
- OTA firmware updates

Cloud Layer Specs

- 99.9% uptime SLA
- AES-256 encryption at rest
- ISO 27001 compliant
- Sub-second dashboard refresh

This architecture keeps processing close to the equipment for speed and resilience, while the cloud layer provides durable storage, centralized intelligence, and secure integrations for downstream business systems.



The Octagonal AIoT Engine

At the heart of Envision 360 is an eight-layer intelligence architecture. Each layer plays a specific role in converting raw signals from your equipment into smart, actionable business decisions — automatically and continuously.



Application Layer

REST API & GraphQL endpoints; role-based access control (RBAC); web & mobile UI with real-time WebSocket feeds



Presentation Layer

D3.js rendering engine; KPI widgets, heatmaps & Sankey diagrams; configurable alert thresholds & report templates



Session Layer

OAuth 2.0 / JWT token management; session timeout policies; multi-tenant isolation with namespace partitioning



Transport Layer

TCP/UDP with QoS prioritization; MQTT v5 broker with persistent sessions; TLS 1.3 end-to-end encryption; retry & ACK logic



Network Layer

IPv4/IPv6 dual-stack routing; SD-WAN failover; LoRaWAN, NB-IoT & LTE-M gateway bridging; VPN tunnel management



Data Link Layer

Modbus RTU/TCP & BACnet MS/TP framing; CRC error detection; RS-485 bus arbitration; Ethernet 802.3 MAC addressing



Meta Data Layer

Asset ontology tagging (ISO 50001 aligned); time-series context enrichment; geospatial & equipment hierarchy mapping



Physical Layer

CT clamps, RTDs, voltage transducers, PIR & flow meters; 50–1000Hz sampling; RS-485, 4–20mA & Pulse output interfaces



These 8 layers work seamlessly together to convert raw sensor data into smart, real-time business decisions — with no manual intervention needed.

The Quadrupole Energy Quotients

Enervision 360 operates across four levels of intelligence, each building on the last to move your operations from reactive to fully optimized. Together, these four quotients turn raw data into decisive business action.



Q1 – Anomaly Quotient

Detects problems the moment they occur, before they escalate.



Q2 – Predictive Quotient

Forecasts equipment failures days or weeks in advance.



Q3 – Preventive Quotient

Schedules smart maintenance to stop issues before they start.



Q4 – Prescriptive Quotient

Prescribes actions that optimize performance and maximize energy savings.



Built for Every Industry



Hotels

Control energy room-by-room. Reduce waste from vacant rooms and optimize HVAC for guest comfort and efficiency.



Office Buildings

Monitor HVAC, lighting, and shared equipment across every floor — cutting costs and improving workplace comfort.



Hospitals

Ensure critical equipment never fails unexpectedly. Maintain compliance and patient safety with continuous monitoring.



Manufacturing

Maximize uptime on production lines. Prevent costly stoppages with predictive insights and automated alerts.

The Business Impact Is Measurable

30–45%

Energy Cost Reduction

Typical savings achieved by optimizing usage and eliminating waste across all connected systems.

30–40%

Maintenance Cost Savings

Reduction in maintenance spend through predictive scheduling and fewer emergency repair callouts.

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Unexpected Breakdowns

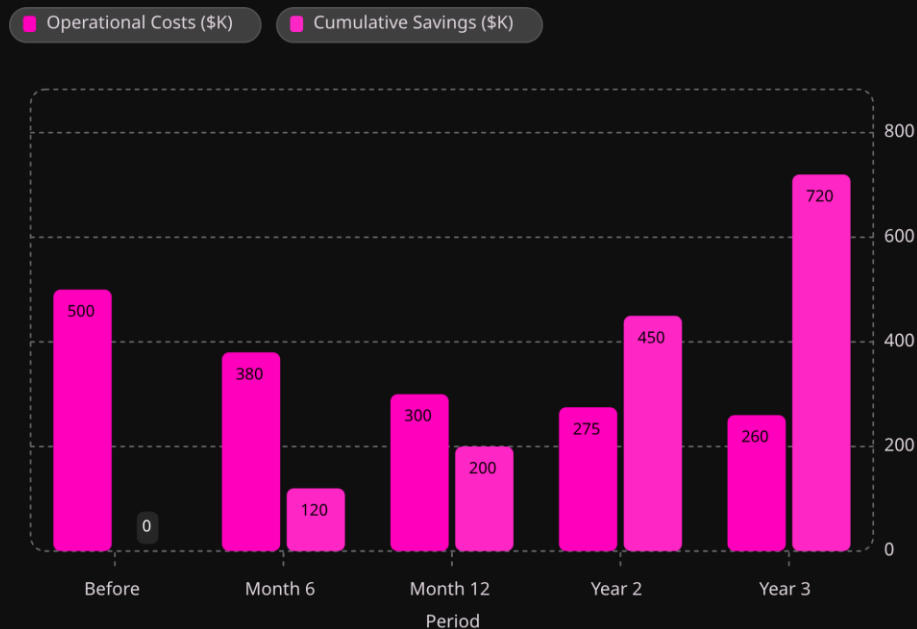
Near-zero unplanned downtime when predictive and preventive quotients are fully operational.

24/7

Operational Visibility

Continuous, real-time visibility across every facility, every system, every hour of the day.

ROI That Speaks for Itself



Payback in ~12 Months

Most organizations recover their full investment within the first year through energy and maintenance savings alone.

Long-Term Value

By Year 3, cumulative savings significantly outpace the initial investment — delivering sustained cost reduction year after year.

Low-Risk Deployment

No major infrastructure overhaul required. Enervision 360 integrates with existing systems quickly and cost-effectively.



What Management Gains

1

Full Visibility Across All Locations

One unified dashboard gives leadership a real-time view of energy use and equipment health at every site — no more blind spots.

2

Better, Faster Decisions

Clear data and plain-language recommendations empower executives to act with confidence — backed by intelligence, not guesswork.

3

Reduced Operational Risk

Early warnings and predictive insights eliminate the element of surprise, protecting your operations and bottom line.

4

Controlled, Predictable Costs

Replace reactive, unpredictable spending with planned, optimized expenditure across energy and maintenance budgets.

What Operations Teams Gain

→ 🚨 Early Alerts

Issues are flagged well before they become failures — giving your team time to plan, not panic.

→ 📅 Less Emergency Work

With failures predicted in advance, costly last-minute repairs and overtime become the exception, not the rule.

→ 📅 Easy Maintenance Planning

Scheduled maintenance windows replace emergency callouts, making workloads predictable and manageable.

→ ⚡ Faster Issue Resolution

Precise diagnostics tell your team exactly what's wrong and where — cutting troubleshooting time dramatically.





Sustainability & ESG Impact

Enversion 360 doesn't just save money — it actively supports your organization's environmental and social governance commitments by driving measurable reductions in energy consumption and carbon output.



Reduced Energy Consumption

By eliminating waste and optimizing usage, facilities consume significantly less energy — translating directly to lower emissions.



Lower Carbon Footprint

Real reductions in electricity usage mean measurable decreases in your organization's carbon output — reportable and auditable.



Supports ESG Goals

Provides the data, reporting, and measurable outcomes that demonstrate genuine commitment to environmental responsibility to stakeholders, investors, and regulators.

A Dashboard Built for Decision-Makers

Designed for Everyone – No Technical Training Required

Envision 360's interface is clean, intuitive, and purpose-built for business users. Whether you're reviewing performance from a boardroom or a mobile phone, the information you need is always one tap away.

- Live energy and equipment monitoring across all sites
- Instant alerts delivered to phone, email, or dashboard
- Automated reports for management review
- Customizable views by site, department, or asset



Summary: Everything Your Organization Needs

Enervision 360 is a complete, scalable solution for organizations that are serious about controlling costs, protecting operations, and building a more sustainable future.



Smarter Energy

Reduce electricity costs by up to 45% through AI-powered optimization and real-time waste elimination.



Smarter Maintenance

Predict failures, plan maintenance proactively, and eliminate costly emergency breakdowns for good.



Smarter Operations

Give every level of your organization — from the C-suite to the operations floor — the visibility they need to perform at their best.



Smarter Growth

Scalable across any number of sites and building types. Grows with your organization as you expand.



Smarter Energy. Smarter Operations.

Enervision 360 gives your organization the intelligence to reduce costs, prevent failures, and operate sustainably — all from a single, easy-to-use platform.

"The most cost-effective decision you'll make this year is the one that prevents the most expensive problem you haven't had yet."

■ [READY TO GET STARTED?](#)